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Advances and Trends in Micro- and Nano-Fabrication

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Abstract: Micro- and nano-fabrication are major impellers of modern industrial and scientific development that allow extremely complex structures to be manufactured at microscopic and nanoscopic dimensions. These technologies include methods like lithography, laser structuring, additive manufacturing, chemical vapor deposition, and self-assembly in the manufacturing of materials and devices that provide enhanced performance and functionality. These technologies are imperative to implement in electronics, biomedical technology, aerospace, and energy storage, where accuracy and miniaturization are of the essence. Artificial intelligence, machine learning, and hybrid manufacturing technologies are also merging to allow scalability, cost-effectiveness, and environmental sustainability of production. Exceptional advances have been made, yet scalability, reproducibility, and the environment await to be exploited. This chapter presents a general introduction of the basics, methods, and applications of micro- and nano-manufacturing, and concise innovation and trend.

Keywords: Micro-manufacturing, nano-manufacturing, lithography, additive manufacturing, laser structuring, nanotechnology, self-assembly, biomedical applications, semiconductors, energy storage, aerospace engineering, sustainable manufacturing.

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1. Introduction

Micro- and nano-manufacturing are two new technologies that came down to us from high-level methodologies used for fabricating ultra-tiny, ultra-accurate structures and devices. Micro- and nano-manufacturing technologies are unstoppable now in the areas of electronics, biotechnology, and materials sciences since these technologies can make functional components with the highest level of precision. One of micro-manufacturing's finest accomplishments has been the application of two-photon polymerization (TPP) technology. The technology can be used by researchers to produce extremely complex three-dimensional structures with unprecedented precision. TPP has transformed the industry by the ability to fabricate intricate shapes that are impossible using conventional methods. In biomedical engineering, for instance, TPP is employed to print personalized scaffolds for tissue engineering, enabling researchers to develop improved medical devices. The technology is also applied in optics and photonics, where it optimizes the performance of micro-optical components. In nano-fabrication, techniques induced by lasers have highly improved additive manufacturing capability [1-3]. The most potent technique is femtosecond laser structuring, which offers precise fabrication of structures at the nano-scale. The technique is commonly applied in manufacturing materials that possess special properties, e.g., pyrolytic carbon, commonly applied in energy storage and medicine. Through employment at this nanoscale, scientists can enhance properties of materials and promote innovation in areas such as energy and electronics [4]. This hybrid approach increases efficiency and minimizes wastage of material as well, which aligns with greater emphasis placed on sustainable processes in manufacturing. Nanoparticle synthesis is increasingly happening for creating drug delivery systems, for which synthesis is gaining rapid advancements as well. Size specification less than 100 nm of nanoparticles poses challenges for nanoparticle production manufacturing processes for PLGA (poly(lactic-co-glycolic acid)), and so many continue to call for standardization. Conventional techniques produce high polydispersity indexes, which create difficulties in the preparation of efficient drug delivery systems [5]. Recent advances in microfluidic devices have shown promising results in overcoming such limitations through providing controlled particle size and distribution, thereby improving the therapeutic effectiveness [6]. As research continues, the ability of the sensors to offer essential information in most areas is growing. The use of micro- and nano-manufacturing methods also includes the production of advanced materials, especially for energy storage. The 3D printing method of producing micro-electrochemical energy storage devices has been of interest due to its potential to develop high-performance batteries and supercapacitors. These devices are availed of the benefit of material engineering at micro and nano scales, resulting in increased energy efficiency and density. Research being conducted for these devices also seeks to overcome the limitations of cost and scalability existing today and be of immediate relevance for market readiness [7,8]. Moreover, nanomedicine is another important subject matter that has found prominence in academia because nanotechnology offers the approach to fabricating targeted drug-delivery devices having enhanced therapeutic outcome and lesser toxicity. Nanoparticle application in drug delivery allows for targeted delivery of the affected tissue, enhancing the effectiveness of treatment in diseases like cancer [1-6]. The capability to design nanoparticles with desirable surface properties also improves their function in drug delivery procedures. Apart from drug delivery, the combined effect of micro- and nano-fabrication technologies in tissue engineering has made it possible to create biomimetic scaffolds that facilitate cell growth and tissue regeneration. With the use of 3D printing technologies such as melt electro writing, scaffolds having a controlled architecture as needed can be fabricated to replicate the natural extracellular matrix [9,10]. The possibility of developing effective filtration systems through the application of nanomaterials points towards the double advantage of enhanced performance along with addressing major environmental issues. With the growth of micro- and nano-manufacturing, the use of artificial intelligence and machine learning will likely be the center of attention in optimizing the manufacturing process. These technologies are capable of maximizing precision, minimizing waste, and maximizing overall efficiency in production lines [11]. The combination of advanced manufacturing methods with AI analysis is able to shape the future of manufacturing, where companies can quickly respond to evolving customer needs. In summary, micro- and nano-manufacturing are some of the advanced technologies that are at the forefront of progression in most industries. The combination of advanced manufacturing methods, in addition to continued research and development, is shaping the way to produce high-performance devices and materials. As scalability, cost, and environmental concerns are alleviated, the potential of these technologies to transform industries grows. The future of micro- and nano-manufacturing holds unprecedented levels of breakthrough that will define the future of technology and technology use in everyday life. Graphene nanomaterials are the basis of modern industries. Besides industrial usage, it is also of major significance as a component of machines utilized there. As a component of electronic devices, graphene nanomaterials are utilized effectively in the formation of p-n junction, active and passive components. Various graphene-based nanomaterial properties have been studied [12-34].

2. Experimental detail

2.1 Principles of Micro- and Nano-Manufacturing

Micro- and nano-manufacturing technology has emerged as the focal point in contemporary manufacturing because of ever-growing needs for precision, efficiency, and the capability to produce highly complex structures at very small dimensions. The technology has evolved extremely fast, and it has propelled the evolution of industries like electronics, medicine, and materials science. In their nature, micro- and nano-manufacturing technologies depend on material processing at very small scales to create devices and structures with superior properties and functionalities. Material processing accuracy is at the core of this practice, and techniques such as laser lithography, electrospinning, and additive manufacturing are important. For example, Lin et al. showed that femtosecond laser structuring is able to produce sub-50 nm features, as required for high-resolution applications [35]. Likewise, thiol-ene click lithography and other nanofabrication methods have tremendous light sensitivity and are able to create nanoscale intricate patterns [36-40]. All of these developments highlight the need for precision in the transformation of material properties at the nanoscale. Nano- and micro-manufacturing combined with new materials has also given rise to new composites and structures. For instance, the inclusion of polydopamine in nano-injection molded interfaces significantly improves adhesive properties and has specific applications for biomedical implants and flexible electronics [41]. Powder metallurgy nanocomposites have also imparted new avenues in material design with enhanced mechanical and thermal properties [42]. This complementarity of advanced materials and manufacturing processes highlights the revolutionary nature of micro- and nano-manufacturing across industries. Micro- and nano- manufacturing's second defining feature is that it can create complex geometries which were unimaginable in the past. Methods like 3D printing and microfluidics have changed the way and extent to which complex parts are designed and produced. As an example, gradient origami metamaterials were created to offer out-of-plane curvatures, which could be used in soft robotics and adaptive structures [43]. Secondly, laser-enabled self-assembly of thin films into micro-rolls is just a sample of the possibility of creating multifunctional devices with novel intrinsic properties [44]. All of these examples demonstrate the richness of micro- and nano-manufacturing in producing complex structures for different purposes.

2.2 Scaling Laws and Material Behavior

This strategy enables enhancements to be made to production processes as well as the predictive modeling of material performance, and therefore more effective and efficient ways of producing. Moreover, the integration of artificial intelligence and machine learning into production processes is exposing opportunities to wiser, more responsive systems that can learn to respond to new conditions in real-time [45]. Sustainability is also a core underpinning principle driving the creation of micro- and nano-manufacturing technologies. The urge for greener production processes has fostered the development of green materials and processes. A notable example is the application of biogenic silver nanoparticles for cancer therapy as an illustration of nanotechnology applicability in sustainable healthcare operations [46]. The application of nano-cellulose composites in wastewater treatment is another such illustration of the viability of applying nanotechnology in environmental processes [47]. These illustrations reflect the increasing awareness of sustainability as a core concept in the design and deployment of micro- and nano- fabrication technologies. Additionally, scaling issues of micro- and nano-fabrication processes continue to be a subject of research. Although numerous methods have been effective at laboratory scales, scaling up such methods to industrial production levels is fraught with numerous challenges in terms of cost, reproducibility, and process control. Recent research has aimed to surmount such limitations by creating hybrid manufacturing methods that synthesize classical and contemporary strategies to scale up efficiency [48]. Additional research in this field is pivotal to unlocking the true potential of micro- and nano-manufacturing across industries. In short, the ideas of micro- and nano-manufacturing are multifaceted, encompassing precision, integration of advanced materials, complex geometries, computational modeling, sustainability, and scalability. As the technologies evolve, they can transform manufacturing processes and enable the development of innovative products in a wide range of applications. Future R&D in the area will, sure enough, shape manufacturing years from now and fuel innovation in materials science, electronics, and health care.

2.3 Scaling Laws and Material Behavior

Scaling laws and material response are fundamental concepts of micro- and nano-manufacturing since they determine how materials react when their size and shape are changed at these smaller scales. The ideas are important in determining how the manufacturing process is optimized and functionality is maintained for micro- and nano-scale devices (Figure 1). Size to mechanical properties response is one of the fundamental scaling laws principles. When materials are reduced to the nano and micro scales, their mechanical properties are different from bulk behaviors because of higher surface-to-volume ratios and prevalence of surface effects. For example, Cheng et al. established a scaling law for the impact toughness of amorphous alloys and showed that the mechanical properties of the material can be influenced considerably by its size Cheng et al. [49]. This finding is valuable for application where impact resistance is critical, e.g., in protective coatings and structural components. Moreover, the material properties at micro and nano scales can lead to strange effects that are not observed with macroscopic counterparts. For instance, Wacker et al.'s study presents the application of ultramicrotomy for imaging and examining 3D structured materials to identify the material's mechanical response at various scales [50]. Based on this multimodal imaging method, researchers can learn how the microstructural constituents impact the performance of the material, thus enabling enhanced design and manufacturing practices. In micro-manufacturing processes, scaling effects can also affect production quality and efficiency. For example, the research by Abeni et al. on micro-milling highlights the influence of tool run-out and its influence on cutting forces, which are critical in realizing the desired surface finishes and dimensional accuracy in micro-components [51]. Miniaturization of machining processes adds complexity that should be well modeled and controlled to achieve desired material properties. It is a process where the deposition technique provides the capability to form intricate structures in terms of target applications, demonstrating the synergy between material properties and manufacturing processes. Moreover, the scaling laws' problems are also applicable in micro/nano device designing where both thermal and mechanical properties need to be taken into consideration carefully. Li et al.'s research on microgroove structure fabrication cites the need to strictly control the fabrication process to obtain smooth surfaces and certain performance parameters [52,53]. This emphasizes the importance of scaling law understanding in micro/nano device fabrication and design since any deviation from predicted behavior can result in severe performance problems. In summary, scaling laws and material behavior are central to the discipline of micro- and nano-manufacturing. The distinctive material properties at these dimensions require complete comprehension of the influence of size on mechanical properties, which guides the creation of novel advanced manufacturing methods and the design of high-performance devices. Study in this field continues to yield new information that will fuel innovation and effectiveness in micro- and nano-manufacturing. Micro- and nano-manufacturing processes involve a wide range of processes that lie at the basis of fabricating parts in nanometer dimensions, especially in electronics, biomedical devices, and advanced materials. Micro- and nano-manufacturing processes are divided into different categories based on their operation principles, material interactions, and applications (Figure 2). Lithography continues to be a pillar of micro- and nano-fabrication, used to pattern materials with great accuracy. Conventional photolithography has been extended by methods like extreme ultraviolet (EUV) lithography and electron beam lithography (EBL) to produce high-resolution patterns needed for the fabrication of semiconductors [54]. Nanoimprint lithography (NIL) is another prospective technique by which it is possible to obtain high-fidelity nanostructure replication, as seen in the production of flexible sensors (Qin et al., 2025).

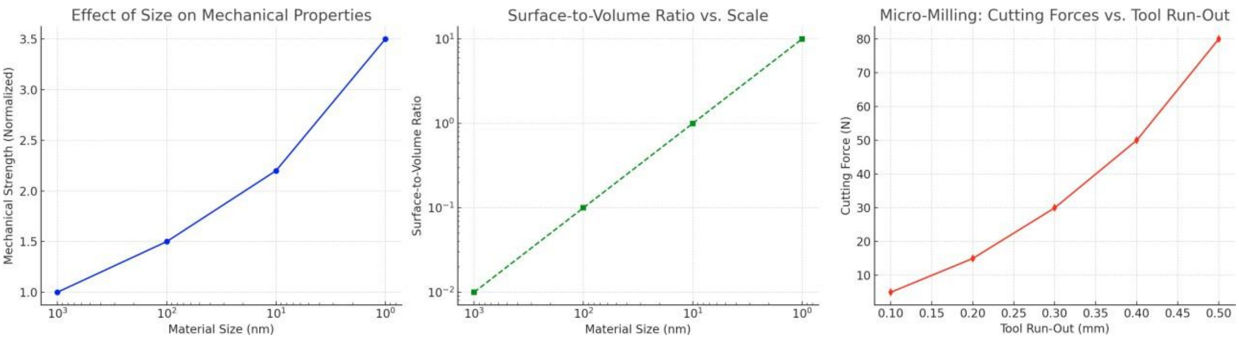

Fig.1. Scaling Effects on Material Behavior in Micro- and Nano-Manufacturing

2.4 Classification of Micro- and Nano-Manufacturing Techniques

These techniques are required for use in processes with complex designs, such as sensors and microfluidic devices. Additive manufacturing (AM) methods, including 3D printing, have transformed the production of components on the micro- and nano-scale. Different AM techniques, including material extrusion, vat photo-polymerization, and binder jetting, enable the creation of complex shapes hard to manufacture using conventional subtractive techniques. An example is micro-LED technology in display panels, where stamp transfer processes enable the accurate deposition of micro-components [55]. Moreover, the use of AM with materials such as elastomers and composites allows the creation of functional structures with pre-defined properties [56]. Subtractive processes like micro-milling and micro-turning are utilized to improve the surface finish and dimensional precision of parts. Micro-milling, for example, is best suited for making complex patterns on materials like metals and polymers that are critical in applications from dental implants to micro-electromechanical systems (MEMS) [57]. The accuracy of these processes is of utmost significance since they have direct bearing on the functionality and reliability of end products. Hybrid manufacturing processes combine multiple techniques to leverage the advantages of each. For example, the integration of additive and subtractive methods allows for the creation of complex parts with high precision and reduced material waste. Techniques such as micro hot embossing are also gaining traction, particularly in the production of microfluidic devices, where the ability to replicate fine features is paramount [58].

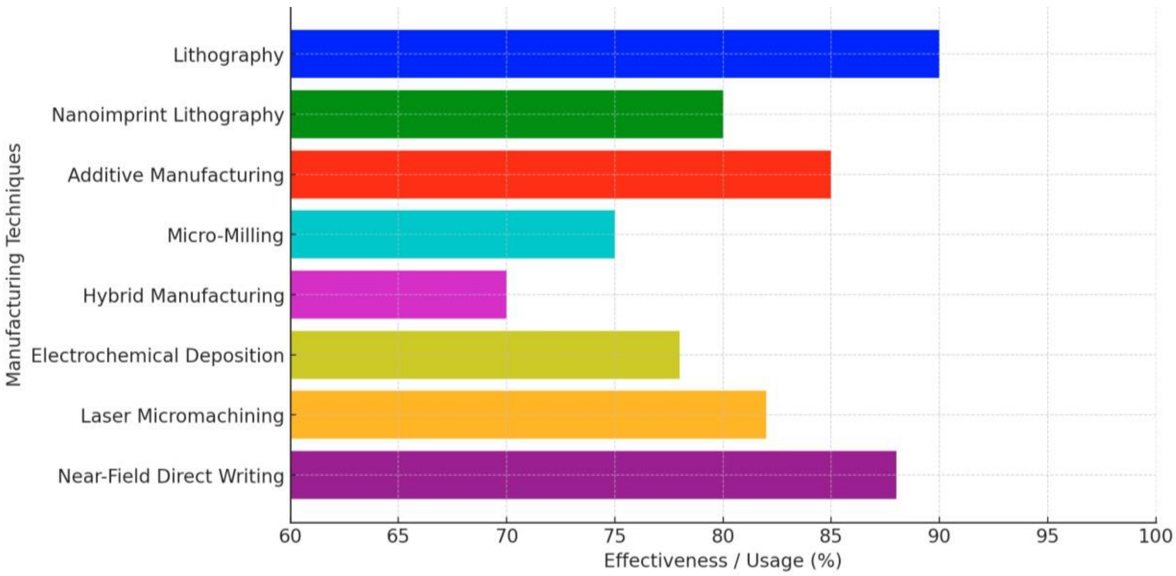

Fig.2. Comparison of Micro- and Nano-Manufacturing Techniques

This approach not only enhances production efficiency but also enables the fabrication of multi-material components. Electrochemical deposition methods, including localized electrochemical deposition, are increasingly used for fabricating microstructures with high aspect ratios and complex geometries. These techniques allow for the precise control of material properties at the micro and nano scales, making them suitable for applications in electronics and sensors [59,60]. The ability to create fine features through electrochemical processes is particularly advantageous for producing components that require intricate designs and high performance. Microfabrication encompasses a range of methods specifically designed for producing micro-scale components. Techniques such as laser micromachining and micro-discharge machining enable the precise cutting and shaping of materials, which is essential for applications in optics and photonics [61]. The advancements in these techniques have led to the development of innovative products, including micro-optical devices and integrated photonic systems. Recent advancements in micro- and nano-manufacturing have introduced novel techniques such as near-field direct writing and droplet-based 3D printing. Near-field direct writing allows for the precise deposition of materials at the nanoscale, enabling the creation of complex lattice structures [62]. Similarly, droplet-based 3D printing offers unique advantages in the rapid production of intricate components, particularly in the fabrication of heterogeneous devices [63]. These emerging techniques are expanding the possibilities for micro-manufacturing, paving the way for new applications and innovations.

3. Micro-Manufacturing Techniques

Lithography is a core micro-manufacturing process used for precise patterning of materials. Conventional photolithography has developed into newer versions like soft film nanoimprint lithography, which is excellent at replicating nanoscale patterns on substrates through flexible molds. The process is especially valuable in semiconductor manufacturing and biopharmaceutical production due to it being economical and extremely accurate in replicating patterns. In addition to this, projection micro-stereolithography (PμSL) has evolved as a potential tool for creating complex three-dimensional structures with improved accuracy, positioning it as an excellent choice in applications involving microfluidics and optics [64,65].

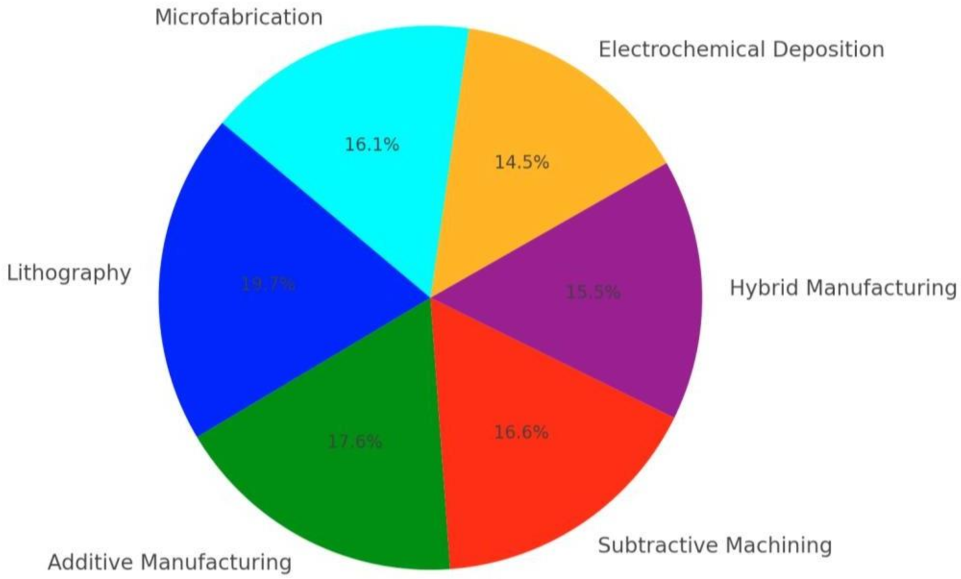

Fig.3. Micro-Manufacturing techniques usage distribution

Additive manufacturing (AM) is a collection of methods that construct parts layer by layer. Selective laser melting (SLM) and fused deposition modeling (FDM) are among the most widely used methods for producing complex microstructures. For example, SLM has been modified to be used for micro-scale manufacturing, allowing one to produce complex geometries that are hard to produce using conventional techniques [66]. The method is beneficial for creating heterogeneous pieces and devices with different material properties (Figure 4). Subtractive machining techniques such as micro-milling and micro-EDM (Electrical Discharge Machining) are to credit for enhancing the surface finish and dimensional accuracy of micro-components. Similarly, micro-EDM is used for drilling hard materials, which are quite difficult to process by conventional means, allowing features of minute size with high precision [67]. Hybrid manufacturing methods are the marriage of different methods to leverage the best they have to offer. Taking additive and subtractive approaches together, for instance, enables one to design intricate parts with high accuracy while losing the least amount of material [68]. Micro hot embossing is also rapidly gaining much attention, especially in the fabrication of microfluidic devices, where subtle details must be duplicated [58]. This hybrid approach enhances production efficiency and enables the production of multi-material parts that meet the needs of a particular application. Electrochemical deposition techniques, such as localized electrochemical deposition, find common usage in the manufacturing of high aspect ratio microstructure with complicated geometry. The techniques offer the ability to manage material characteristics with nanoscale and microscale precision and suit for usage in sensors and electronics [67]. Fabricating detailed features through electrochemical operations offers distinct value when manufacturing components needing complicated layouts and superior performance. Microfabrication involves various methods that are specifically tailored to fabricate micro-scale components. The precision cutting and shaping of materials using techniques like laser micromachining and micro-discharge machining are used to manufacture components used in optics and photonics devices (Wacker et al., 2023). With the developments of these processes, new innovative products like micro-optical components and photonic integrated systems are manufactured. Recent micro-manufacturing advancements brought forth new methods including near-field direct writing and electrohydrodynamic atomization. Near-field direct writing enables the deposition of material in high precision at the nanoscale level that can be used to build up complex lattice structures [62]. These new technologies are opening up new opportunities for micro-manufacturing, which is leading the way to new applications and innovation.

3.1 Lithography-Based Methods

Lithography methods are the major methods of micro-manufacturing, which enable the precise patterning of material in the nano and micro ranges. The methods have their significance in a range of applications, including semiconductor production, biomedical devices, and microfluidic devices. The remainder of this paper presents an introduction to key lithography methods, their fundamentals, and current progress in the field. One of the most popular lithographic processes is photolithography, and it utilizes light to project a pattern onto a substrate with a light-sensitive layer (photoresist). The process is to expose the photoresist to ultraviolet (UV) radiation using a photomask that determines the desired pattern to be created. Among the most recent innovations in photolithography are breakthroughs in extreme ultraviolet (EUV) lithography that allow for feature generation in the sub-5 nm regime, improving the resolution and performance of semiconductor fabrication Giannopoulos et al. (2024) [69]. Further, techniques like grayscale lithography have been explored to develop intricate three-dimensional structures, increasing the use of photolithography in many industries [70]. Nanoimprint lithography is a technique that entails the mechanical deformation of a mold with nanoscale features into a thin resist film on a substrate. The approach enables replication of complex patterns with high fidelity and works extremely well for the manufacturing of large-area nanostructures. NIL has found application in photonic devices and sensors in recent research, having been adequately illustrated in its capability and versatility of micro-manufacturing [71]. In addition, improvements in NIL techniques, such as the use of polymer masters produced by laser ablation, improved the scalability and precision of the process [72]. Direct laser writing is mask less lithography that involves the use of focused laser beams to polymerize or ablate materials to create complex microstructures. This technique is most suitable for rapid prototype construction and the production of personalized devices. Utilization of femtosecond lasers that can be used to create high-resolution structures with minimal thermal abuse to the surrounding material [73] has been the recent advancement in DLW. DLW has also been employed in combination with two-photon polymerization methods for the fabrication of three-dimensional microstructures with complex geometries [74]. Electron beam lithography uses a focused electron beam to write unique patterns into an electron-sensitive resist-coated substrate. The process has unmatched resolution, and thus it is appropriately suited to be applied in very fine feature applications. EBL is commonly used in research and development environments to prototype and manufacture intricate nanostructures [75,76]. Recent developments have witnessed an increase in the throughput and reduced cost of the EBL method, making it viable for different applications [77]. The capability of the method to create periodic structures without masks is a cost advantage of the method over conventional lithography methods. Soft lithography encompasses a set of techniques using elastomeric stamps to pattern substrates. It is most suitable in the fabrication of microfluidic devices and biomaterials since it can replicate finer features precisely. Recent focus has been on the fabrication of novel stamp materials and novel pattern transfer procedures so that more complex structures can be fabricated [78]. Advanced research has established the viability of grayscale lithography in the production of functional surfaces with controlled features, expanding its use in micro-manufacturing [79]. Micro-manufacturing as a whole includes lithography-based methods with its accuracy and diversity needed in use today. With ongoing advancements toward those technologies, they continue to get stronger in what it can do in terms of more complex functional microstructures across industries.

3.2 Mechanical Micro-Machining

Mechanical micro-machining is a core manufacturing discipline that addresses material removal in a controlled way to achieve micro-scale components. The process is of significant application in aerospace, medical equipment, and electronics sectors where precision and intricate designs are required. The following sections describe an overview of dominant techniques and features of mechanical micro-machining from the literature. Micro-milling flexibility enables it to machine various materials, including metals and polymers, and is thus suited for various applications. Furthermore, digital twin technology has also been utilized in the management of tool wear in micro-milling, enhancing the process efficiency and reliability [80]. Micro-turning is another significant mechanical micro-machining process, particularly in turning and machining high-precision cylindrical components. Micro-turning is a process involving the utilization of high-speed spindles and high-precision tooling on some micro-lathes to satisfy the needs of tolerance. Micro-turning has been realized, through studies, to be effectively utilized in machining hard materials, for instance, titanium alloys that have wide application in aerospace engineering [81-83]. Micro-turning requires optimization of cutting parameters for minimum tool wear and good surface finish (Huang et al., 2023).

Micro-grinding utilizes small grinding wheels to remove material in micro quantities. Micro-grinding is ideal to obtain good surface finish and the attainment of close tolerances. Micro-grinding has been the way to get improved surfaces in difficult materials based on research data that makes micro-grinding an optometry application technique and a precision engineering process [83-85]. Improvements in high-performance grinding equipment and techniques, such as the use of diamond abrasives, have also enhanced micro-grinding performance [86]. Micro-electrical discharge machining (μEDM) is an unconventional machining process that removes material from a workpiece through electrical discharges. The process is extremely useful in machining hard work material and intricate profiles, which cannot be machined through conventional methods. Research in μEDM has further emphasized lately to optimize process parameters to obtain enhanced machining efficiency and surface integrity [87-90]. The ability to machine micro-features such as micro-holes and cavities makes μEDM a very important manufacturing process in micro-component manufacturing [91]. Tool wear is still one of the dominant problems in mechanical micro-machining since it has a direct impact on the quality of machined components. Studies have shown that built-up edge (BUE) in machining can lead to increased wear and worse surface finishes [92]. Tool wear management through advanced techniques like machine learning and digital twin technology erases such issues and generally improves the performance of the micro-machining process [80]. Mechanics-based micro-machining techniques are increasingly being employed in numerous applications such as biomedical devices, micro-electromechanical systems (MEMS), and microfluidics. The requirement for accurate miniaturized components remains among the highest on the list of drivers for micro-machining technology. Amongst some of the upcoming trends can be the use of novel materials, i.e., composites and ceramics, in micro-machining processes and hybrid machining methods which integrate traditional with non-traditional techniques for attaining superior performance [81-83].

3.3 Additive Micro-Manufacturing

Additive micro-manufacturing is disruptive manufacturing technology used to make complex micro-scale components in the fields of electronics, biomedical devices, and high-performance materials. The process, otherwise referred to as 3D printing, provides for production layer by layer of complex geometry parts that are not possible with conventional subtractive manufacturing. There are numerous technologies within additive micro-manufacturing that have specialized capabilities and applications. Two-Photon Polymerization (TPP) is a sub-nanometric 3D printing technique involving intensive pulses of laser beams polymerizing photosensitive resin, used in micro-optics, biomedical scaffolds, and nanostructured materials. Micro-Stereolithography (Micro-SLA) utilizes standard SLA techniques to manufacture small complex components with the surface finish of smoothness, normally used to fabricate microfluidic devices and precise biomedical components. Electrohydrodynamic Printing (EHDP) is able to offer ultra-fine jetting of functional ink to create high-resolution microscale features with great contribution towards flexible electronics, sensors, and biomedical devices. Selective Laser Sintering (SLS) and Selective Laser Melting (SLM) are two laser energy-driven processes for sintering or melting powder particles into intricate microstructures and thus have application in aerospace lightweight and high-strength parts and medical implants. Continuing innovation in additive micro-manufacturing is providing new innovations to many industries that enable highly optimized, tailored, and miniaturized products. Applications of the next generation will be unveiled by subsequent research that continues to improve the accuracy, scalability, and material scope of these technologies.

- Stereolithography (SLA) - SLA is a widely used technique that employs a UV laser to cure liquid photopolymer resin into solid structures. This method is known for its high resolution and ability to produce complex geometries with smooth surface finishes. Recent advancements in SLA technology have enabled the printing of high-precision components, showcasing its versatility in various applications [93];
- Selective Laser Melting (SLM - SLM is a powder bed fusion technique that uses a laser to selectively melt and fuse metallic powders layer by layer. This method is particularly effective for producing high-density components with complex internal geometries. Research has shown that SLM can be used to manufacture intricate parts, demonstrating its potential in precision engineering [94];
- Two-Photon Polymerization (TPP) - TPP is a high-resolution additive manufacturing technique that utilizes focused laser beams to initiate polymerization at the nanoscale. This method allows for the creation of intricate microstructures, such as photonic devices and microfluidic channels, with sub-micrometer resolution. Recent studies have highlighted the potential of TPP for fabricating functionalized optical devices, expanding its application in sensing and communication technologies [95];

Applications of Additive Micro-Manufacturing The applications of additive micro-manufacturing are vast and varied, spanning multiple industries:

- Biomedical Devices - Additive manufacturing techniques are increasingly being utilized to produce patient-specific medical devices, such as implants and prosthetics (Figure 4). The customization capabilities of additive manufacturing allow for the creation of components that closely match individual anatomical structures, enhancing the effectiveness of medical treatments [96].

- For instance, the use of 3D printing for fabricating tissue scaffolds has shown promise in regenerative medicine [97];
- Microelectronics - The miniaturization of electronic components has driven the adoption of additive manufacturing in the electronics industry. Techniques such as EHD printing and TPP enable the fabrication of high-resolution micro/nano-scale structures that are essential for the development of advanced electronic devices, including sensors and circuit components [98-99];
- Energy Storage - Additive manufacturing is also making strides in the field of energy storage. The ability to create hierarchical micro/nanostructured electrodes through 3D printing has been shown to enhance the performance of rechargeable batteries and supercapacitors, leading to improved energy density and efficiency [100-102];
- Challenges and Future Directions - Despite the significant advancements in additive micro-manufacturing, several challenges remain. Issues related to material properties, process optimization, and the integration of multiple materials in a single print are areas that require further research and development. For instance, the mechanical properties of additively manufactured components can be influenced by factors such as layer thickness and material orientation, necessitating careful control during the printing process [103];
- Future directions for additive micro-manufacturing involve exploring new materials, including biocompatible polymers and next-generation ceramics, to expand applications. The integration of artificial intelligence and machine learning techniques in process design and optimization of additive manufacturing is also anticipated to improve efficiency and lower cost of production [104].

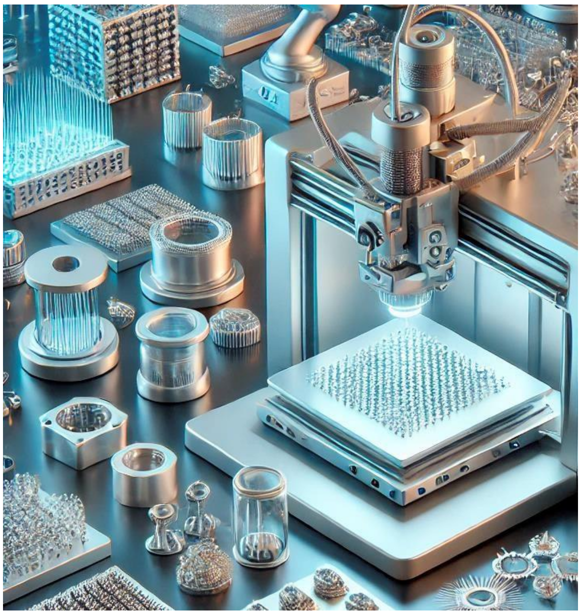

Fig.4. Advanced Micro-Scale Additive Manufacturing System

4. Nano-Manufacturing Techniques

Nano-fabrication methods constitute the core of nanoscale material and device manufacturing, the foundation of technological progress in electronics, medicine, and materials science. The methods enable the fine-tuning of material properties and functions and hence the realization of novel technologies across a broad spectrum of applications. A few of the most important nano-fabrication methods, their principle of operation, and recent advancements in the field are discussed in detail below:

1. Nanoimprint lithography is a high-resolution patterning technique that involves mechanically pressing a mold with nanoscale features into a thin layer of resist on a substrate. This method allows for the replication of intricate patterns with high fidelity and is particularly effective for producing large-area nanostructures. Recent advancements in NIL have focused on improving the materials used for molds and the techniques for pattern transfer, enabling the fabrication of more complex structures. NIL has shown promise in applications such as semiconductor manufacturing and biophotonics due to its cost-effectiveness and scalability [105].
2. Femtosecond laser processing is a technique that employs ultrafast laser pulses to modify materials at the micro and nanoscale. This method allows for precise material removal and structuring without significant

thermal damage to the surrounding material. Recent advancements in femtosecond laser processing have enabled the fabrication of micro/nano-architectures with unique optical properties, making it suitable for applications in sensors and photonic devices. The ability to create complex structures with high spatial resolution positions femtosecond laser processing as a leading technology in nano-manufacturing [106].

3. EHD printing is a novel additive manufacturing technique that uses electric fields to deposit materials in micro/nano-scale patterns. This method offers significant advantages in terms of scalability and cost-effectiveness, making it suitable for the fabrication of electronic components and sensors. Recent research has demonstrated the ability of EHD printing to produce high-resolution patterns with minimal material waste, positioning it as a disruptive technology in the field of nano-manufacturing [107,108]. The versatility of EHD printing allows for the integration of various materials, including conductive inks and polymers, further expanding its application range.
4. Chemical vapor deposition is a widely used technique for producing thin films and nanostructures by chemically reacting gaseous precursors to form solid materials on a substrate. CVD is particularly effective for creating high-quality semiconductor materials and coatings. Recent advancements in CVD processes have focused on improving the uniformity and scalability of film deposition, enabling the production of nanostructured materials with tailored properties for applications in electronics and energy storage [109]. The ability to control the deposition parameters allows for the precise tuning of material characteristics, making CVD a vital technique in nano-manufacturing.
5. Self-assembly processes utilize the propensity of molecules to self-assemble into structured patterns on their own without aid. This process is especially useful for the synthesis of nanostructures and material with pre-defined functionalities. Modern advances in self-assembly processes have been geared towards increasing the degree of control over the process of assembly to facilitate intricate nanostructures with pre-defined properties for utilization in drug delivery and sensors [110]. Self-assembly capability in forming hierarchical structures provides new windows for designing multiframe materials of high performance.

4.1 Chemical and Physical Vapor Deposition (CVD & PVD)

Physical Vapor Deposition (PVD) and Chemical Vapor Deposition (CVD) are two of the most utilized methods of the coating and thin film deposition on different materials with numerous uses from electronics to material science and optics. Each one has their own advantages and based on the required properties and objectives of the desired material and applications; each process is utilized separately:

- Chemical Vapor Deposition (CVD) - CVD is a process that involves the chemical reaction of gaseous precursors to form a solid material on a substrate. The process typically occurs at elevated temperatures and can produce high-quality thin films with excellent uniformity and adhesion. CVD is particularly advantageous for depositing materials that require precise control over composition and thickness. Recent advancements in CVD technology include the development of plasma-enhanced CVD (PECVD), which allows for lower deposition temperatures and improved film quality Asgartooran et al. PECVD is particularly useful for depositing dielectric films and coatings on temperature-sensitive substrates, making it suitable for applications in microelectronics and optoelectronics. Furthermore, CVD techniques have been successfully applied to create coatings with enhanced mechanical properties, such as titanium carbide (TiC) coatings, which exhibit excellent wear resistance [111,112].
- Physical Vapor Deposition (PVD) - PVD is a vacuum-based coating technique that involves the physical transfer of material from a solid source to a substrate. The material is vaporized and then condensed onto the substrate surface, forming a thin film. PVD techniques include sputtering, evaporation, and laser ablation, each with its specific advantages. Sputtering is a widely used PVD method that involves bombarding a target material with energetic ions, causing atoms to be ejected and deposited onto the substrate [113]. Evaporation is another common PVD technique that involves heating a material in a vacuum until it vaporizes and then allowing it to condense on the substrate. This method is particularly effective for depositing metals and alloys, and it is often used in the production of optical coatings and reflective surfaces. The advantages of evaporation-based PVD techniques include higher purity of the film and better control of the film thickness [114].
- Comparison of CVD and PVD - While both CVD and PVD are effective for thin film deposition, they differ in their mechanisms and the types of materials they can deposit. CVD is generally preferred for materials that require chemical reactions to form, such as silicon dioxide or silicon nitride, while PVD is often used for metals and alloys. Additionally, CVD processes typically operate at higher temperatures compared to PVD, which can be a limiting factor for temperature-sensitive substrates [115].

Both the PVD and the CVD processes are commonly found in various industrial processes. CVD finds application in the semiconductor in the deposition of conductive films as well as insulating films, whereas metal interconnects and barrier layers make use of PVD [116].

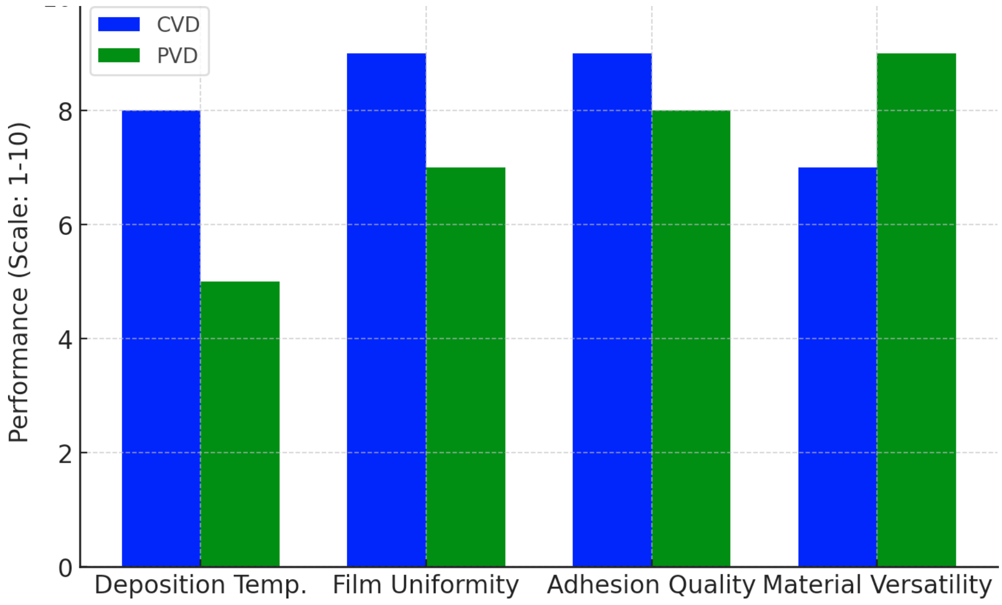

Fig.5. Comparison of Chemical Vapor Deposition (CVD) and Physical Vapor Deposition (PVD) Based on Key Parameters

In optics also, both PVD as well as CVD are used in the formation of anti-reflective coating, as well as high-refractive index mirrors. In addition, the two processes find application in cutting tool protective coatings production, improving their wear properties and lifespan [117].

4.2 Nanolithography and Patterning Techniques

Nanolithography and patterning processes play a significant role in the development of nanoscale devices and structures, paving the way for innovation in biotechnology, photonics, and electronics. Nanolithography and patterning processes allow for the possibility of control of material property and function at the nanoscale, giving rise to innovation in applications from sensors to drug delivery systems. In the sections below, we introduce some notable nanolithography and patterning processes, mechanisms, and recent developments. Nanoimprint lithography is a recent technique for micro/nano-scale large-area patterning by a mold printing patterns on a resist-coated substrate. It is inexpensive and simpler compared to conventional photolithography. This method has proved to possess high potential in the high-resolution patterning application of semiconductor devices and photonic structures. Femtosecond laser writing describes a direct-write method that involves the application of ultra-fast laser pulses for direct inscription of nanostructures within material. Direct femtosecond laser writing facilitates the creation of intricate three-dimensional nanostructures with high precision accuracy and minimal thermal damage. Recent evidence favors direct femtosecond laser writing in the fabrication of Si-Cr nano-alloys beneficial for use in lithography-free direct surface patterning necessary in metallic met surfaces [118]. The technique's flexibility makes it eligible for applications in optics and materials science. Dip-pen nanolithography is achieved by the deposition of material on a substrate using a pointed tip with the aid of capillary force. One is able to arrange biomolecules and nanoparticles at the nanoscale with high accuracy. It has been used in the recent past to pattern hydrophobic and hydrophilic gold and magnetite nanoparticles, showing its capability to generate intricate nanostructures for sensor and catalysis purposes [119]. It is the capability of controlling such minuscule scales during the process of deposition that makes DPN a useful nanofabrication tool. Laser interference lithography employs the interference pattern generated by the superposition of the laser beams to pattern periodic structures onto a substrate. The technique is applied in the fabrication of high-precision, large-area patterns with uniformity. Advances in LIL have enabled the development of multiscale pattern structures by combining nanotransfer printing and laser micromachining to be used in photonic devices and sensors. LIL is an economically less costly technique compared to traditional lithography techniques because it has the ability to produce periodic structures without the need for masks. Electrospinning lithography is a technique which combines electrospinning and lithographic techniques to fabricate nanofibrous structures in an intended pattern. Electrospinning lithography can be employed to create intricate hierarchical microstructures by iteratively repeating the process involving microcontact patterning and electrospinning [120,121]. Recent research has proven the viability of the technique to be applied in tissue engineering and filtration, where the improved properties of nanofibers can be leveraged. Block copolymer lithography takes advantage of the self-assembly of block copolymers to generate nanoscale patterns. The method allows the generation of well-ordered nanostructures whose sizes can be controlled. Current advancements in the field have focused on the use of directed self-assembly techniques to enhance the control of the generated patterns, allowing the fabrication of intricate nanostructures for electronics and photonics [122]. The ability to produce high-fidelity large-area patterns positions block copolymer lithography in an effective set of nanofabrication techniques. Nanosphere lithography uses self-assembled monolayers of nanospheres as a template to create patterns. Highly reproducible and uniform nanostructures in periodic form are created by nanosphere lithography. The application potential of the use of nanosphere lithography in the fabrication of functional nanostructures has been investigated in recent studies, and it has proved useful for catalysis and biosensing [123]. Defects are still hard to control in the self-assembly. Ease and simplicity of scalability of this technique render it a favorable choice for mass-scale nanofabrication.

4.3 Self-Assembly and Molecular Manufacturing

Self-assembly and molecular manufacture are central principles of nanotechnology that allow the production of complex materials and structures by spontaneous self-organization of molecules. These methods rely on non-covalent forces for the formation of organized structures, which can be exploited in many various applications, including drug delivery, biosensing, and materials science. Here we summarize some essential findings of recent studies of self-assembly and molecular manufacture methods: Mechanisms of Self-Assembly - Self-assembly process is initiated by different non-covalent forces, such as hydrogen bonding, van der Waals forces, π-π stacking, and electrostatic forces. These forces facilitate spontaneous molecule assembly into highly ordered structures. For example, Zheng et al. showed that even though some molecules are very simple, their self-assembly processes can be intricate and uncertain when placed in contact with metal surfaces Zheng et al. [124]. This unpredictability is indicative of the necessity for sophisticated modeling methods, including graph neural networks, to make molecular configurations predictable from experimental data. Amphiphilic Molecules and Hydrogel - Development Amphiphilic molecules with hydrophilic and hydrophobic segments have a tendency to self-assemble into diverse structures when brought into contact with an aqueous environment. Yue et al. studied co-self-assembly of amphiphiles in the form of nanocomposite hydrogels and concluded that they can be developed with more desirable mechanical properties [125]. Modulation of hydrogel morphology and function through self-assembly provides new promise for drug delivery and tissue engineering applications. DNA Self-Assembly - DNA nanotechnology has been a topic of strong interest because it can create complex nanostructures via programmed self-assembly. For instance, Sun et al. used optimized protocols for DNA self-assembly by chaotropic agents to reduce unwanted random aggregates and increase the yield of the desired structures [126,127]. In addition, Cai and Lin also stressed the application of DNA origami as a multi-purpose platform for the fabrication of nanoscale devices with drug-delivery and biosensing capabilities. Due to the programmability of DNA, the assembly process is under complete control, and complex nanostructures can be designed. Peptide Self-Assembly - Peptides are another type of biomolecules that possess self-assembly characteristics, creating nanostructures with targeted properties. Mohanty et al. synthesized pathway-controlled multicomponent peptide hydrogels, illustrating that the pathways of self-assembly are essential in dictating the formed material properties [128]. Such peptide material control for targeted functionalities is important for drug delivery and regenerative medicine applications. Supramolecular Chemistry - Supramolecular self-assembly refers to the self-assembly of molecules into macroaggregates by non-covalent interactions. Co-assembly versus peptide amphiphile self-sorting was investigated by Sangji et al. and revealed that multimeric routes of self-assembly could lead to different nanostructures with specific functionalities [129]. This work is a perfect example of the importance of knowing the thermodynamic and kinetic factors that govern supramolecular assembly. Applications in Nanotechnology - Self-assembly and molecular manufacturing applications are diverse. Similarly, Xiao et al. utilized supramolecular self-assembly for designing electrodes for intracellular recording, illustrating the use of such methodologies in biosensing [130]. Even with the improvement in self-assembly methods, controlling the assembly process and achieving reproducibility is not easy. Temperature, solvent conditions, and concentration of molecules also contribute significantly to the outcome of self-assembly. Research has to develop strict methods of managing these parameters in order to make self-assembled structures more consistent. Short of it, self-assembly and molecular manufacturing are at the heart of the process of developing nanotechnology to produce complex devices and materials with designed properties. Further research is expanding frontiers and potential applications and provides hope for revolutionary application in many varied fields [131].

5. Applications of Micro- and Nano-Manufacturing

Micro- and nano-manufacturing techniques have a wide array of applications across various fields, including electronics, medicine, energy, and materials science. These advanced manufacturing methods enable the creation of intricate structures and devices at scales that were previously unattainable. Below, we synthesize key applications of micro- and nano-manufacturing, supported by relevant literature (Figure 6).

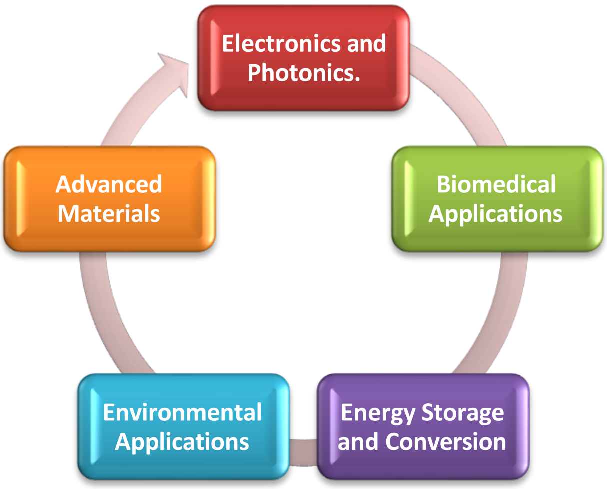

Fig.6. Key Applications of Micro- and Nano-Manufacturing Techniques

5.1 Electronics and Photonics.

Micro- and nano-manufacturing techniques are crucial in the electronics industry, particularly in the fabrication of semiconductor devices, sensors, and photonic components. For instance, the use of nanoimprint lithography (NIL) has been shown to enhance the performance of micro-light-emitting diodes (micro-LEDs) by enabling the precise patterning of nanoscale features, which is essential for improving brightness and efficiency Wang et al. [38].

5.2 Biomedical Applications

In the biomedical field, micro- and nano-manufacturing techniques are employed to create devices and materials for drug delivery, tissue engineering, and diagnostics. For example, Xu et al. demonstrated the fabrication of micro/nano dual needle structures using two-photon polymerization, which holds potential for applications in microfluidics and targeted drug delivery systems. Furthermore, the development of amino-acid-encoded supramolecular nanocarriers has shown promise for enhancing drug delivery efficiency and targeting specific cells. The ability to engineer materials at the micro and nanoscale allows for the creation of scaffolds that mimic natural tissues, promoting cell growth and regeneration [71,131].

5.3 Energy Storage and Conversion

Micro- and nano-fabrication technologies are also essential in the progress of future energy conversion and storage materials. Zhong et al. highlighted the advantages of silicon anode materials with micro/nano-structure for lithium-ion batteries with better performance than general materials (Zhong et al., 2024). The capability to fabricate high-energy-density electrodes through techniques such as chemical vapor deposition (CVD) and physical vapor deposition (PVD) enhances the energy storage device's efficiency and shelf life. Nanostructured material applications in supercapacitors have also demonstrated remarkable improvements in charge-discharge rates and energy density [132,133].

5.4 Environmental Applications

Micro- and nano-fabrication technologies have central roles in environmental applications, especially in sensor and filtration system design. For example, the production of superhydrophobic surfaces from micro/nano-structured materials has been shown to improve water filtration system efficiency by avoiding fouling and increasing flow rates. Apart from that, applications of nanomaterials in catalytic converters and air filters have been very promising in cleaning pollutants and enhancing air quality [81-83].

5.5 Advanced Materials

The possibility of designing materials at the micro- and nanoscale has changed the creation of composite materials with highly specific properties. Among the new trends in the field is the production of nanostructured films using Physical Vapor Deposition (PVD) and Chemical Vapor Deposition (CVD) technologies. These deposition techniques allow the production of materials with superior mechanical strength, enhanced thermal stability, and enhanced chemical resistance, which are extremely suitable for aerospace, biomedical devices, and electronics applications. Additionally, the latest advances in femtosecond laser processing have given a powerful tool to produce complex nanostructures with superior feature control. With this process, surface property can be engineered at the nanoscale and one can produce materials with engineered optical properties. This has enormous applications in photonics, sensing, and imaging, where light needs to be manipulated at the nanoscale. These technological advances serve to emphasize the increasing significance of nanoscale engineering in materials science. By the refinement of deposition processes and laser processing, scientists are continually pushing the boundaries of what can be achieved when designing materials with performance and functionality beyond anything previously known. As the need for high-performance materials continues to grow across all industries, continued research in this area will become increasingly important to the future of material design and use [134].

6. Robotics and Automation

Micro- and nano-fabrication techniques are increasingly being used in micro/nanorobot manufacturing for medical applications. Micro- and nano-fabrication techniques can be used for performing extremely precise actions at the cellular level, e.g., cell delivery and on-demand manipulation of cells. The latest advancements in the design and fabrication of micro/nanorobots have demonstrated that they can be immensely helpful in a wide range of applications, including diagnostics and treatment. The integration of these technologies with artificial intelligence is likely to enhance the performance and functionality of micro/nanorobots in the future. Micro- and nano-manufacturing technology pervades and has diverse applications in numerous industries and applications. From electronics and energy storage to biomedicine devices and environmental applications, micro- and nano-manufacturing technologies provide manufacturing capability for producing new materials and devices with enhanced properties and performance and functional benefits. Increased growth in the field of new technology and applications to this sector is dependent on continuous research and development work, and that leads to further technological breakthroughs in certain fields [135-137].

6.1 Electronics and Semiconductors

Micro- and nano-manufacturing technologies have transformed the semiconductor and electronics industry to the extent of facilitating small, efficient, and highly functioning devices. Micro- and nano-manufacturing technologies facilitate advanced structures and materials necessary for application in modern-day electronics. This article provides a review of the recent literature regarding key findings of micro- and nano-manufacturing applications within the semiconductor and electronics industries.

6.1.1. Organic Electronics

Organic field-effect transistors (OFETs) are a good example of micro-manufacturing technology, especially for the fabrication of flexible and light electronic devices. Huang et al. showed that laser mapping annealing can be used to realize high-performance OFETs using liquid crystalline organic semiconductors Huang et al. [94]. The capability of controlling the microstructure of organic semiconductors using self-assembly methods is highly important for optimizing device performance.

6.1.2. Layered Materials and MXenes

Layered materials like MXenes have been of interest because of their peculiar electrical characteristics and the possibility of their application in electronics. Song et al. investigated the self-assembly of MXene films in layered form, emphasizing the modularity of the surface work function by chemical modification [138]. Modularity of this type is at the core of electronic device functionality optimization, e.g., sensors and energy storage devices.

6.1.3. Stretchable Electronics

Yet another field where micro- and nano-manufacturing methods have helped fuel growth is in wearable electronics. Sun et al. developed the air/liquid interfacial self-assembly method to prepare intrinsically stretchable films, which can be applied in wearable technology [126]. The process showcases the capability of creating flexible and resilient electronic elements that resist mechanical deformation.

6.1.4. Semiconductor Membrane

Exfoliation Chang et al. characterized semiconductor membrane exfoliation technology and applications and provided the opportunity for the production of light and thin electronic devices [139]. Such an action, normally governed by chemical etching and peeling in mechanics, forms a core activity in the construction of future electronic devices that are less heavy in weight and more efficient.

6.1.5. Colloidal Self-Assembly

Colloidal self-assembly processes have also been employed in designing ultrathin metasurfaces for use in optics and photonics devices. Jiang et al. demonstrated that self-assembly with colloids had the capability of being utilized in making metasurfaces for complete visible range absorption, showing the scalability and affordability of such a process [140]. The possibility of constructing advanced optical devices with the capability of employing self-assembly in the generation of intricate nanostructures provides new avenues.

6.1.6. Self-Assembled Monolayers (SAMs)

Self-assembled monolayers (SAMs) are finding widespread application in semiconductor devices to alter surface characteristics and enhance charge transport. Dickson et al. provided an overview of the capability of SAMs made from poly (ionic liquid) block copolymers to maximize semiconductor device performance [141]. The capacity to design surface characteristics through SAMs underpins device efficiency and functionality optimization.

6.1.7. Quantum Dots and Nanostructures

Quantum dots (QDs) are semiconductor crystals with average electronic and optical properties at the nanometer scale. Wu et al. employed self-rolled-up ultrathin single-crystalline silicon nanomembranes as on-chip tubular polarization photodetectors to verify the benefit of employing QDs in high-end photonic devices [67]. The employment of QDs improves the device efficiency and offers new possibilities for miniaturization.

6.1.8. Nanoparticle Assembly

Self-assembly of nanoparticles is a strong method for the fabrication of functional materials with precise properties. Cossio et al. explained scalable self-assembly of nano and microparticle monolayers through aerosol-assisted deposition and highlighted its application for the production of high-resolution patterns for electronic devices [142]. With this method, one can create intricate structures with the ability to control composition and positioning precisely.

6.1.9. Flexible and Transparent Electronics

The demand for flexible and transparent electronic devices has promoted the evolution of micro- and nano-fabrication techniques. Chen et al. demonstrated the use of dry-transferable photoresists for conformal patterning in ultrathin flexible electronics, highlighting the importance of such techniques in next-generation device fabrication [87-90]. Low-cost high-performance flexible electronics are required in wearable technology as well as smart device integration. Micro- and nano-fabrication technology applications in electronics and semiconductors are wide-ranging and diverse, encompassing organic electronics, stacked material, stretchable electronics, and complex nanostructures. Research and development in this area continues to pave the way, and the possibility of creating smaller, efficient, and very functional electronic devices is achievable. With advancements in technology, future semiconductor and electronics manufacturing will be revolutionized.

6.2 Biomedical and Healthcare Applications

Micro- and nano-fabrication techniques have transformed the biomedicine and healthcare industries to manufacture next-generation devices and materials to improve diagnostics, therapy regimens, and regenerative medicine. The overview presents important applications and current trends in the field based on current research. The most promising development is the use of self-assembled peptide hydrogels as drug delivery systems. The hydrogels are highly biocompatible and can deliver therapeutic drugs efficiently [143]. Sedighi et al. are of the opinion that such material is useful in targeted drug delivery, wound healing, and 3D cell culture scaffolds. Their effectiveness is also reinforced by the ability of these materials to respond to specific stimuli and hence release the drugs in controlled amounts at the targeted site. This ability helps to increase the effectiveness of therapy with minimal side effects. Micro- and nano-fabrication techniques are also crucial in tissue engineering. Jiang et al. (2024), for instance, explored the use of biodegradable nanobowls as delivery tools for cargo and bioimaging probes, which are of significant value in tissue regeneration. Because of their concave shape, the nanobowls are capable of encapsulating and delivering bioactive molecules effectively, and hence they are highly valuable reagents in regenerative medicine. The second most important development is in diagnostics, and self-assembled nanostructures enhance biosensor efficiency. Sarkar et al. [144] outlined the development of peptide-based biosensors that take advantage of the advantage of self-assembly to offer sensitivity and specificity for biomolecule detection. The biosensors are extremely affine and selective and may potentially be excellent weapons for early diagnosis and disease monitoring. Coatings Self-assembled materials are under investigation for their antibacterial activity, particularly in the case of medical implants and devices. Surface functionalization using self-assembled nanostructures presents new avenues to improve the biocompatibility and shelf life of medical devices. Nanostructured materials formed via self-assembly are also applied in diagnostics and imaging. Zhao et al. documented the use of gold nanoparticles in imaging and targeted therapy and showed their utility to enhance contrast in imaging techniques [145,146]. The tunability of the optical properties of these nanoparticles makes them most applicable for medical therapeutic and diagnostic tracing applications. Stimuli-responsive materials are being used more and more in biomedical applications because they possess the ability to respond to the environment by changing properties. Hou and Liu discussed the development of pH-sensitive peptide aggregates that could be employed for tumoral microenvironments targeting drug delivery and imaging [147]. Therapeutic-delivery materials employed for the delivery of therapeutics after the mentioned adjustment in pH are of most use when targeting tumoral microenvironments. Nanofibers in Regenerative Medicine Electrospun nanofibers have been researched thoroughly to be used in regenerative medicine. Stoica et al. suggested the application of electrospun nanofibers for tissue engineering and drug delivery due to their ability to engineer surface area and high porosity [148]. These features make the nanofibers highly versatile in scaffolding architecture for cell growth and tissue regeneration. The nanostructure possesses the modularity of accommodating more than a single therapeutic moiety, which possesses the ability to offer a customized treatment regimen. Micro- and nano-fabrication sciences are transforming biomedical and healthcare technologies by enabling the fabrication of highly complex materials and devices for the next-generation diagnostics, therapeutics, and regenerative medicine. The advancement in research and development in self-assembly and molecular manufacturing is further pushing the science ahead with new horizon opportunities of applications and better results for patients.

6.3 Aerospace and Automotive Industry

Micro- and nano-fabrication methods have become more central in aerospace and automobile production to facilitate the production of high-performance parts to satisfy specific specifications on function, power, and weight. The technology applied in their production takes into account intricate designs and materials that advance functionality as well as performance of airplanes and vehicles. We summarize below major applications and advancements in micro- and nano-fabrication within these industries from evidence sources. Automotive and aerospace industries are always seeking to reduce weight to improve fuel economy and performance. Micro- and nano-scale production methods such as additive manufacturing allow for easier production of light lattice structures that are strong and minimize material use. For instance, Shaik et al. examined additive manufactured micro-lattice structures in the military industry and identified their energy absorption and crash behavior properties [149]. These materials are especially useful in aerospace, where weight needs to be minimized as much as possible. The development of advanced materials such as carbon fiber-reinforced polymers (CFRPs) and titanium alloys is of significant significance in the development of aircraft and automobile parts. Kariuki et al. synthesized CFRPs' uses and properties, whose corrosion resistance and strength-to-weight ratio were very adaptable to be utilized in aircraft and automobile parts [150]. In addition, with the addition of nanomaterials to these composites, additional improved mechanical properties and strength may be attained to enable them for high-performance use [151]. Micro-machining techniques such as micro-milling and micro-EDM are applied for the production of sophisticated features in aerospace and automobile parts. Accuracy of dimensions as well as surface integrity in drilling micro-holes in titanium alloys was mentioned as an example by Deepu et al. (2023) widely used for aerospace parts. High-quality surfaces as well as the ability to achieve precision geometries is required in order to attain critical component performance as well as reliability. Nano-electromechanical systems (NEMS) and micro-electromechanical systems (MEMS) are used extensively in the aerospace and automobile sectors, mainly for sensing and actuation. Micro-manufacturing technologies spurred miniaturization of the technology, where sensors and actuators can be made as compact as possible to fit into. Liu et al. [152] documented ultra-sensitive MEMS capacitive accelerometers, which have been used widely in the aerospace navigation and control system. Miniaturization of these devices allows complex functions to be embedded in light structures and improve system performance as a whole. Surface treatment technologies like laser surface treatment and micro-arc oxidation are used to enhance the wear and corrosion resistance of components that are working in harsh environments. Mandala et al. studied core material optimization in sandwich structures for the aerospace sector with respect to crash energy absorption [149]. The use of new surface treatments and coatings can potentially add great value to the life and lifespan of components exposed to harsh environments. Additive manufacturing has changed the rapid prototyping of the automotive and aerospace sectors by the capacity to evolve designs quickly and test new ideas. Having the capacity to produce intricate geometries in a matter of time shortens development at a lower cost. (Nguyen and Thomas, 2023) discussed how additive manufacturing offers benefits in hybrid rocket component design, pointing out its significance to today's propulsion systems. The design and manufacturing flexibility is of primary importance in meeting the changing demands of these sectors. Furthermore, micro- and nanofabrication methods have broad applications in energy harvesting and future battery technology.

7. Challenges and Future Directions

The integration of functional devices into functional systems is one of the largest challenges in micro- and nano-fabrication. Future studies should be directed towards the development of novel materials to fulfill particular requirements of particular uses, from biocompatibility to conductance and mechanical strength. Lin et al.'s focus reminds us that the creation of implantable devices, like memristors, must meet stringent regulatory and certification requirements for medical devices [153]. Human use safety must be assured through new materials and devices. Future research must encompass coordination among researchers, regulatory agencies, and industry players to create transparent guidelines for the development and testing of new technologies. Micro- and nano-manufacturing process complexity is a challenge in attaining consistent quality and reproducibility. Future study must aim for the identification of mechanisms to enable process manufacturing optimality for improved productivity and lesser errors, using machine learning and artificial intelligence perhaps as a technique of monitoring and controlling processes. The environmental impacts of manufacturing are increasingly of concern. According to Zhong, electronics and medical applications in 3D printing need to be sustainable [154]. Future trends need to emphasize the creation of sustainable materials and processes with minimal waste generation and energy needs, in accordance with global sustainability demands. The capability to define materials and structures at the microscale and nanoscale is important in realizing their properties and performance. Future studies need to enhance and advance characterization methods with the capability of delivering information regarding the structure-property relationships of emergent materials. Micro- and nano-manufacturing challenges require the sophistication of a multidisciplinary approach on the part of material scientists, engineers, and physicians. For Dou et al., the boundary between electronics and biomedical applications is propelling the development of healthcare technology [137]. Upcoming research will have to concentrate on stimulating cross-disciplinary research collaboration to hasten the creation of new solutions and applications. Micro- and nano-manufacturing technologies have huge potential for propelling several industries, mainly electronics and healthcare. For future prosperity with these technologies, though, issues of integration, material constraint, regulation, and sustainability have to be addressed. Through such a focus, researchers and industry stakeholders are able to provide the platform for new applications and enhanced performance in micro- and nano-manufacturing.

8. Conclusion

Micro- and nano-fabrication has now transformed the modern fabrication process to allow very precise components to be produced to be used in applications such as medicine, electronics, and production. Highly advanced processes such as two-photon polymerization, femtosecond laser fabrication, and self-assembled nanocomposites continue to advance the material science and device engineering. Further, the use of artificial intelligence and automation is progressively enhancing output efficiency and inspection quality. Even with such developments, cost factors, scalability, and environmental effects are primary issues. Solutions to these issues will encompass new-generation methods such as hybrid manufacturing techniques, creation of green materials, and improvement in nanofabrication processes. Micro- and nano-manufacturing will be a primary focus in the future to fuel innovation in fields like biotechnology, electronics, aerospace, and renewable energy industries, designing the future of high-tech solutions.

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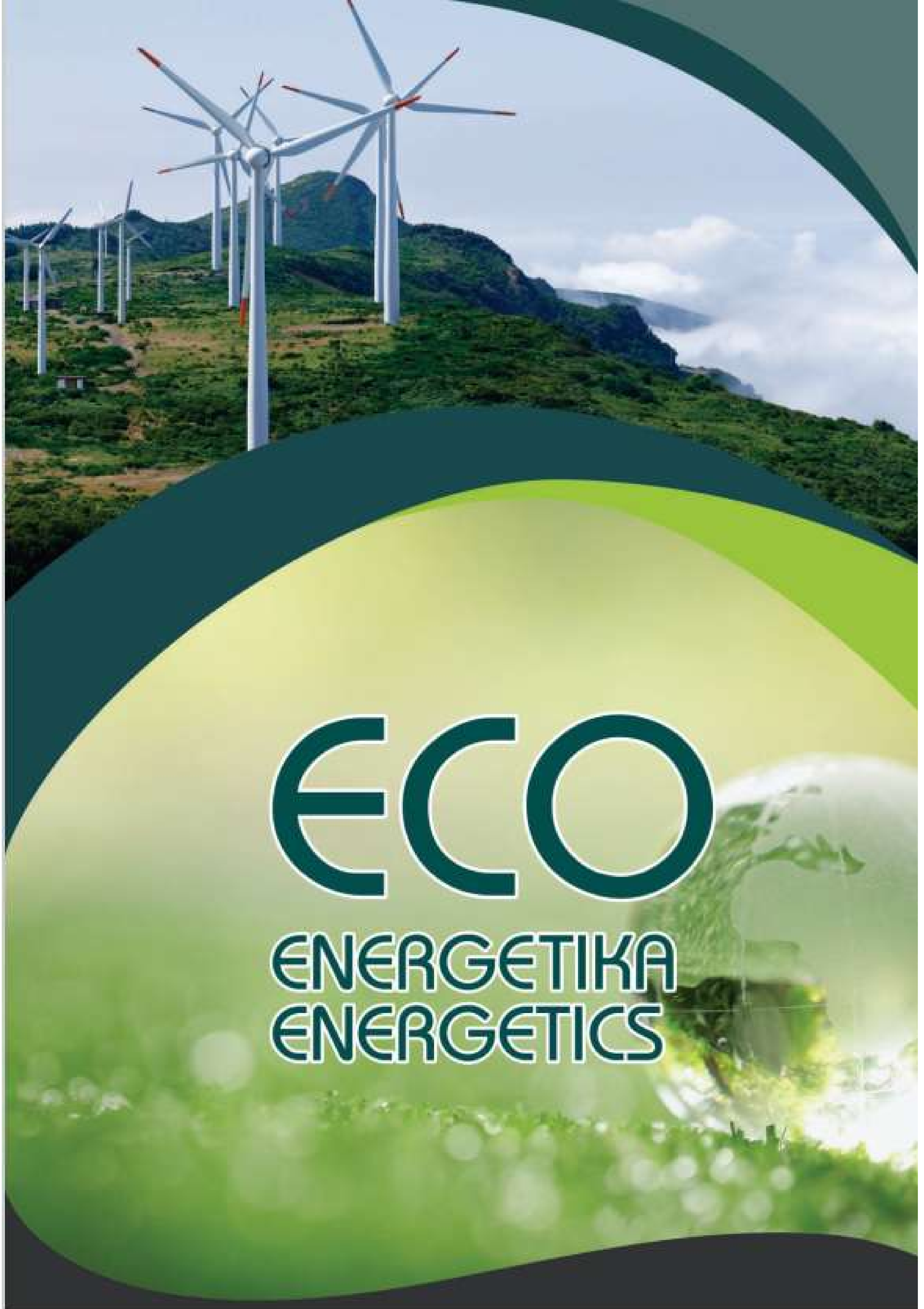
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